

204 Fractal Boundary Condition

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Abstract

The mode functions $\phi_{n,l,j}$ from Doc. 202 (Part II) hold for a simple torus with $D_f = 3$. The fractal dimension $D_f = 3 - \xi_0$ of the FFGFT torus leads to a correction of first order in ξ_0 . This correction modifies the normalisation and the dispersion relation, but leaves the basic structure of the wave functions unchanged. The masses still arise as eigenvalue residues after almost complete compensation.

.1 Starting point: simple torus ($D_f = 3$)

On the simple torus T^3 with side length L_0 the standard integration measure applies:

$$d\mu_0 = d^3x, \quad \int_{T^3} d\mu_0 = L_0^3 = V \quad (1)$$

The mode functions (Doc. 202):

$$\phi_{n,l,j}^{(0)}(t, \mathbf{x}) = \frac{1}{\sqrt{V}} e^{i\mathbf{k}_{n,l} \cdot \mathbf{x}} \cdot Y_{l,j}(\theta, \varphi) \cdot e^{-im_{n,l,j}t} \quad (2)$$

are exactly normalised: $\int_{T^3} |\phi^{(0)}|^2 d\mu_0 = 1$.

.2 The fractal measure

Definition

On the FFGFT torus with fractal dimension $D_f = 3 - \xi_0$ the integration measure is modified. For a fractal medium with Hausdorff dimension D_f the scaled measure applies:

$$d\mu_{D_f} = \left(\frac{|\mathbf{x}|}{L_0}\right)^{D_f-3} d^3x = \left(\frac{|\mathbf{x}|}{L_0}\right)^{-\xi_0} d^3x \quad (3)$$

Expansion for small ξ_0

Since $\xi_0 = 4/30000 \ll 1$, we expand:

$$\left(\frac{|\mathbf{x}|}{L_0}\right)^{-\xi_0} = 1 - \xi_0 \ln \frac{|\mathbf{x}|}{L_0} + \mathcal{O}(\xi_0^2) \quad (4)$$

The fractal measure is therefore:

$$d\mu_{D_f} = \left(1 - \xi_0 \ln \frac{|\mathbf{x}|}{L_0}\right) d^3x + \mathcal{O}(\xi_0^2) \quad (5)$$

.3 Corrected normalisation to first order

The normalisation integral

The corrected normalisation condition reads:

$$\int_{T^3} |\phi_{n,l,j}|^2 d\mu_{D_f} = 1 \quad (6)$$

With the ansatz $\phi_{n,l,j} = N_{n,l,j} \cdot \phi_{n,l,j}^{(0)}$:

$$\begin{aligned} N_{n,l,j}^{-2} &= \int_{T^3} |\phi^{(0)}|^2 d\mu_{D_f} \\ &= \int_{T^3} \frac{1}{V} \left(1 - \xi_0 \ln \frac{|\mathbf{x}|}{L_0}\right) d^3x \end{aligned} \quad (7)$$

Calculation of the correction term

The integral of the correction term:

$$I_{\text{frak}} = -\frac{\xi_0}{V} \int_{T^3} \ln \frac{|\mathbf{x}|}{L_0} d^3x \quad (8)$$

On the cubic torus $[0, L_0]^3$, $|\mathbf{x}|$ is the distance from the origin. The integral yields the mean of the logarithm:

$$\frac{1}{V} \int_{T^3} \ln \frac{|\mathbf{x}|}{L_0} d^3x = \langle \ln(r/L_0) \rangle. \quad (9)$$

In general $\langle \ln r \rangle \neq \ln \langle r \rangle$. Numerically (Monte Carlo) one finds for $[0, L_0]^3$ with distance from the origin $\langle r \rangle = 0.961 L_0$, $\langle \ln(r/L_0) \rangle = -0.094$; for the torus distance (centred) $\langle r \rangle = 0.480 L_0$, $\langle \ln(r/L_0) \rangle = -0.787$. Hence:

$$N_{n,l,j}^{-2} = 1 + \xi_0 \langle \ln(r/L_0) \rangle + \mathcal{O}(\xi_0^2) \approx 1 + 0.69 \xi_0 \quad (10)$$

The coefficient $0.69 = |\ln 0.5|$ used here and below inserts a mean relative distance $c_0 \approx 0.5$ into the logarithm; it follows from the integral in neither of the two readings. With 0.094 or 0.787 one would obtain $N = 1 - 6.3 \cdot 10^{-6}$ or $1 - 5.2 \cdot 10^{-5}$ instead of $1 - 4.6 \cdot 10^{-5}$.

Corrected normalisation factor

Up to first order in ξ_0 :

$$N_{n,l,j} = 1 - \frac{0.69}{2} \xi_0 + \mathcal{O}(\xi_0^2) \approx 1 - 4.6 \times 10^{-5} \quad (11)$$

The correction is tiny ($\sim 10^{-5}$), consistent with the fact that $\xi_0 \ll 1$.

.4 Corrected dispersion relation

The fractal Laplace operator

On the fractal torus the Laplace operator is modified. For small ξ_0 :

$$\Delta_{D_f} = \Delta_3 + \xi_0 \cdot \delta\Delta + \mathcal{O}(\xi_0^2) \quad (12)$$

The correction operator $\delta\Delta$ arises from the modification of the measure:

$$\delta\Delta\phi = -\nabla\left(\ln\frac{|\mathbf{x}|}{L_0}\right) \cdot \nabla\phi = -\frac{\mathbf{x}}{|\mathbf{x}|^2} \cdot \nabla\phi \quad (13)$$

Eigenvalue correction

The corrected Klein–Gordon equation:

$$[\square + \xi_0 \cdot \delta\square] \phi_{n,l,j} = -m_{n,l,j}^2 \phi_{n,l,j} \quad (14)$$

For the plane wave $e^{i\mathbf{k}\cdot\mathbf{x}}$, $\delta\Delta$ yields to first order:

$$\langle\phi^{(0)}|\delta\Delta|\phi^{(0)}\rangle = -\left\langle\frac{\mathbf{x}}{|\mathbf{x}|^2} \cdot \nabla\right\rangle = -ik\langle\cos\theta_{kx}/r\rangle \quad (15)$$

On the torus this term averages to zero, since $\langle\mathbf{x}/|\mathbf{x}|^2\rangle = 0$ (symmetry).

The eigenfrequency correction comes from the scalar part:

$$\delta\omega_{n,l,j}^2 = -\xi_0 \cdot k_{n,l}^2 \cdot \left\langle\ln\frac{|\mathbf{x}|}{L_0}\right\rangle \approx 0.69 \xi_0 \cdot k_{n,l}^2 \quad (16)$$

with the same coefficient 0.69, which does not follow from the integral (Section 3.2); with $|\langle\ln(r/L_0)\rangle| = 0.094$ or 0.787 the respective value would appear here.

.5 The corrected mode functions

The complete mode functions up to first order in ξ_0 :

$$\phi_{n,l,j}^{(D_f)}(t, \mathbf{x}) = \frac{N_{n,l,j}}{\sqrt{V}} e^{i\mathbf{k}_{n,l}\cdot\mathbf{x}} \cdot Y_{l,j}(\theta, \varphi) \cdot e^{-im_{n,l,j}^{(D_f)} t} \quad (17)$$

with:

$$N_{n,l,j} = 1 - 4.6 \times 10^{-5} + \mathcal{O}(\xi_0^2) \quad (18)$$

$$\left(m_{n,l,j}^{(D_f)}\right)^2 = k_{n,l}^2 (1 + 0.69 \xi_0) + \langle\hat{V}\rangle \quad (19)$$

The fractal correction $0.69 \xi_0 \approx 9.2 \times 10^{-5}$ to the kinetic energy does not change the origin of mass qualitatively: the masses remain eigenvalue residues after almost complete compensation; the correction appears only in the fifth decimal place.

.6 Physical significance

Why the correction remains small

The fractal correction is of order $\xi_0 \approx 1.3 \times 10^{-4}$. This is consistent with the mass hierarchy: the particle masses are of order $\xi_0^{p_i} \cdot v_i$, i.e. themselves suppressed by powers of ξ_0 . The wave-function correction lies at the same order.

Orthogonality is preserved

Since the correction (18) is mode-independent (the same N for all (n, l, j)), orthogonality is preserved:

$$\int_{T^3} \phi_{n,l,j}^{(D_f)*} \phi_{n',l',j'}^{(D_f)} d\mu_{D_f} = \delta_{nn'} \delta_{ll'} \delta_{jj'} \quad (20)$$

The non-closure theorem (Doc. 193) holds unchanged.

Completeness

The picture of mass generation from Doc. 202 (Part V) remains correct:

$$m_i^2 = (2\pi n/L_0)^2 \cdot (1 + 0.69 \xi_0) + \langle \hat{V} \rangle \approx (2\pi n/L_0)^2 + \langle \hat{V} \rangle$$

The correction $(1 + 0.69 \xi_0)$ is numerically irrelevant compared with the compensation over 32–39 orders of magnitude.

Summary

The fractal boundary condition arising from $D_f = 3 - \xi_0$ modifies the mode functions of Doc. 202 to first order:

- Normalisation factor: $N = 1 - 4.6 \times 10^{-5}$
- Dispersion correction: $+0.69 \xi_0$ to the kinetic energy

Both corrections are of order $\xi_0 \approx 1.3 \times 10^{-4}$ and do not change the physical structure. The coefficient 0.69 is not derived from the averaging integral; depending on the definition of distance, that integral gives 0.094 or 0.787 ($N = 1 - 6.3 \cdot 10^{-6}$ or $1 - 5.2 \cdot 10^{-5}$). The task from Doc. 202, Problem 2, is thereby formally completed.

Notation

Symbol	Meaning
$D_f = 3 - \xi_0$	Fractal dimension of the FFGFT torus
$\xi_0 = 4/30000$	Fundamental geometric constant
$d\mu_0 = d^3x$	Standard measure (simple torus, $D_f = 3$)
$d\mu_{D_f}$	Fractal measure ($D_f = 3 - \xi_0$)
$\phi_{n,l,j}^{(0)}$	Unperturbed mode functions ($D_f = 3$)
$\phi_{n,l,j}^{(1)}$	First-order perturbative correction in ξ_0
$N_{n,l,j}$	Normalisation factor (corrected)
$N \approx 1 - 4.6 \times 10^{-5}$	Numerical value of the normalisation correction
$\delta\Delta$	Correction to the Laplace operator due to D_f
$\delta\omega_{n,l,j}^2$	First-order eigenfrequency correction
$0.69 \xi_0$	Dispersion correction ($\approx 9.2 \times 10^{-5}$)
$\phi_{n,l,j}^{(D_f)}$	Corrected mode functions up to $\mathcal{O}(\xi_0)$
$m_{n,l,j}^{(D_f)}$	Corrected mass up to $\mathcal{O}(\xi_0)$

References: Doc. 180 (ξ_0), Doc. 193 (non-closure), Doc. 202 (mode functions, origin of mass).